

Comparison Report:
Standard SNN Atan ($\theta = 1.0$) vs SNN Atan ($\theta = 0.4$)

ByzFL SNN Benchmark

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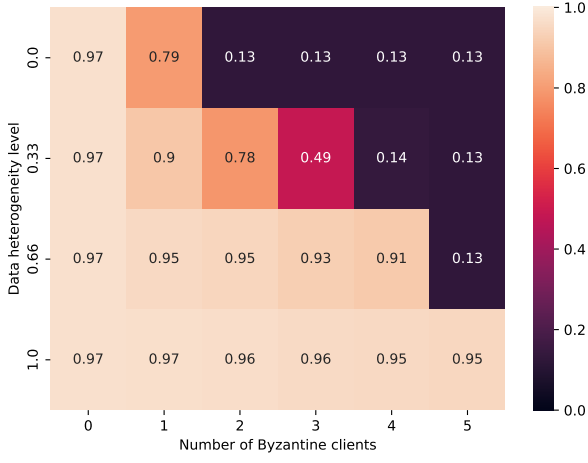
1 Introduction

This report compares the robustness of the standard SNN Atan configuration (spiking threshold $\theta = 1.0$, surrogate gradient sharpness $\alpha = 1.2$) against the same architecture with a lowered spiking threshold ($\theta = 0.4$). We compare the worst-case test accuracy across four robust aggregators (Centered Clipping, Geometric Median, Multi-Krum, and Trimmed Mean) under three different attacks:

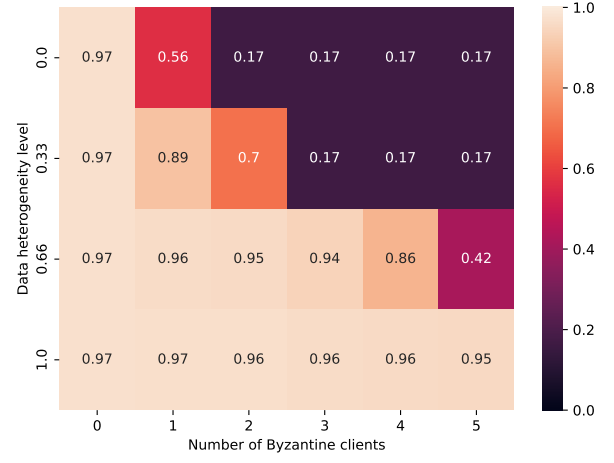
1. Optimal A Little Is Enough (OptALIE)
2. Optimal Inner Product Manipulation (OptIPM)
3. Sign Flipping

2 Worst-Case Robustness (Across All Attacks)

The following figures show the worst-case test accuracy across all three attacks.



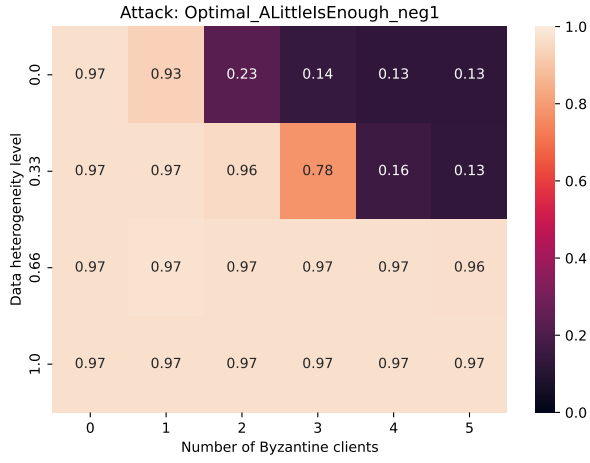
(a) SNN Atan ($\alpha = 1.2, \theta = 1.0$)



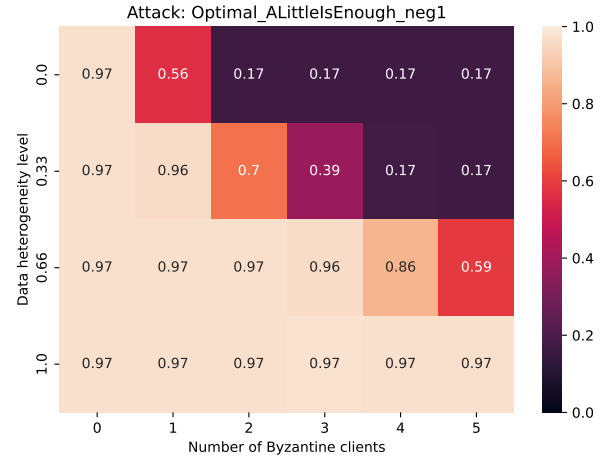
(b) SNN Atan ($\alpha = 1.2, \theta = 0.4$)

Figure 1: Worst-case test accuracy across all attacks.

3 Robustness under Optimal ALIE



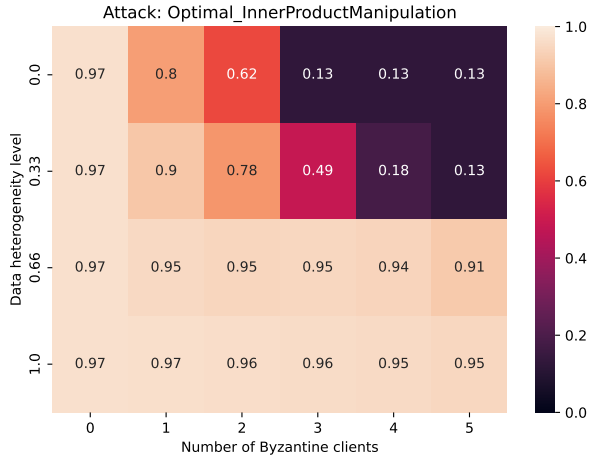
(a) SNN Atan ($\alpha = 1.2, \theta = 1.0$)



(b) SNN Atan ($\alpha = 1.2, \theta = 0.4$)

Figure 2: Test accuracy under Optimal A Little Is Enough (OptALIE) attack.

4 Robustness under Optimal IPM



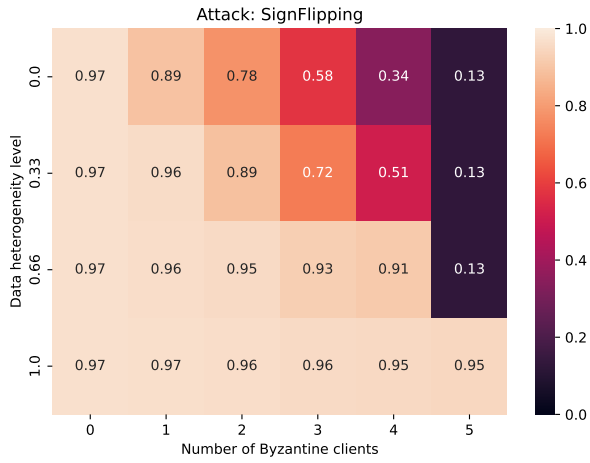
(a) SNN Atan ($\alpha = 1.2, \theta = 1.0$)



(b) SNN Atan ($\alpha = 1.2, \theta = 0.4$)

Figure 3: Test accuracy under Optimal Inner Product Manipulation (OptIPM) attack.

5 Robustness under Sign Flipping



(a) SNN Atan ($\alpha = 1.2, \theta = 1.0$)



(b) SNN Atan ($\alpha = 1.2, \theta = 0.4$)

Figure 4: Test accuracy under Sign Flipping attack.